Baseline model

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1 Overview

This document describes a two-stage stochastic optimization model that decides (i) which candidate cities a robotaxi operator should enter and (ii) how many vehicles to deploy in each entered city, under uncertain robotaxi demand represented by a finite set of scenarios.

The model trades off four cost components:

- fixed city-entry costs,
- per-vehicle costs,
- a penalty for unused service capacity (over-supply), and
- a penalty for unserved robotaxi demand, i.e. foregone contribution margin (under-supply).

Evaluation period. All demand, throughput, and cost quantities are measured over *the same evaluation period*. The model determines the city/vehicle deployment to be applied during that period.

2 Model

2.1 Objective

$$\min_{y,q,O,U} \quad \underbrace{\sum_c F_c y_c}_{\text{entry costs}} + \underbrace{\sum_c K_c q_c}_{\text{vehicle costs}} + \frac{1}{|S|} \sum_{s\in S} \sum_{c\in C} \Big[\underbrace{c_c^+ O_c^s}_{\text{unused capacity}} + \underbrace{c_c^- U_c^s}_{\text{foregone margin}} \Big].$$

The superscripts in c_c^+ and c_c^- denote *excess* and *shortage*, not the sign of the cost. Scenarios are weighted equally ($1/|S|$).

2.2 Constraints

$$\begin{aligned}
&0 \le q_c \le \bar{q}_c\, y_c && \forall c \in C,\\
&\textstyle\sum_c q_c \le Q,\\
&\kappa_c q_c - M_c p_c^s = O_c^s - U_c^s && \forall c \in C,\ s \in S,\\
&y_c \in \{0,1\}, \quad q_c \in \mathbb{Z}_{\ge 0},\\
&O_c^s,\, U_c^s \ge 0.
\end{aligned}$$

Constraint	Meaning
$0 \le q_c \le \bar{q}_c y_c$	No vehicles in a city that is not entered; up to the city-specific limit if it is entered.
$\sum_c q_c \le Q$	Total fleet limit.
$\kappa_c q_c - M_c p_c^s = O_c^s - U_c^s$	Service capacity minus robotaxi demand equals unused capacity minus unmet demand.

2.3 Demand construction

Each city has a pool of potential ride-hailing usage opportunities M_c, which is split into robotaxi demand and non-robotaxi demand:

$$D_c^s = M_c\, p_c^s, \qquad D_c^{N,s} = M_c\,(1 - p_c^s).$$

D_c^s is robotaxi choice demand *before* any vehicle shortage is applied. $D_c^{N,s}$ covers human-driven taxis, other modes, and giving up the trip. In this baseline model, M_c is assumed to be fixed.

2.4 Outcome variables

Given D_c^s and service capacity $\kappa_c q_c$, the following quantities (both in trips) follow from the deployment outcome and do not need to be estimated separately. Here $(x)^+ = \max\{x, 0\}$.

$$O_c^s = (\kappa_c q_c - D_c^s)^+ \quad \text{(unused capacity)}, \qquad U_c^s = (D_c^s - \kappa_c q_c)^+ \quad \text{(unserved demand)}.$$

3 Notation

Symbol	Meaning	Type and unit
C	Set of candidate cities	Input set; cities
S	Set of demand scenarios	Input set; scenarios
y_c	1 if city c is entered, else 0	Decision variable; binary
q_c	Number of vehicles deployed in city c	Decision variable; vehicles (integer)
κ_c	Trips one vehicle can serve during the period	Parameter; trips/vehicle per period
Q	Total vehicles available for deployment	Parameter; vehicles
$\bar{q}_c$	Upper bound on deployment in city c	Parameter; vehicles
F_c	Fixed cost of city entry and basic infrastructure	Parameter; currency per period
K_c	Per-vehicle preparation, deployment, and holding cost	Parameter; currency/vehicle per period
c_c^+	Additional loss per unit of unused capacity	Parameter; currency/unused trip
c_c^-	Foregone contribution margin per unit of unserved demand	Parameter; currency/unserved trip
M_c	Potential ride-hailing usage opportunities in city c	Parameter; opportunities per period (data or assumption)
p_c^s	Probability of choosing the company's robotaxi	Scenario input; dimensionless, $[0, 1]$
D_c^s	Robotaxi choice demand, $M_c p_c^s$	Derived input; trips per period
$D_c^{N,s}$	Non-robotaxi demand, $M_c(1 - p_c^s)$	Derived input; trips per period
O_c^s	Unused service capacity	Outcome variable; trips per period
U_c^s	Unserved robotaxi demand	Outcome variable; trips per period

Notes.

- Demand inputs are M_c and p_c^s; in the baseline model M_c is fixed and uncertainty enters only through p_c^s.
- Outcome variables (O_c^s, U_c^s) are computed from demand and deployment, not predicted from separately collected data.

4 Key assumptions and open points

- A single evaluation period must be fixed; all parameters are expressed over it.
- p_c^s must be specified relative to reference service conditions (e.g. fares, waiting times).
- M_c is not directly observable; it is constructed from observed-usage patterns and assumptions.
- Scenarios are equally likely.